

Google Cloud Food Delivery Analytics Pipeline

Project Overview

This project demonstrates an end-to-end cloud analytics solution built using Google Cloud Platform (GCP). The objective was to transform raw food delivery transaction data into meaningful business insights through data engineering, cloud warehousing, and business intelligence reporting.

The project simulates a real-world food delivery company that receives daily order transactions and requires a scalable analytics solution to monitor operational performance, customer satisfaction, and delivery effectiveness.

Business Scenario

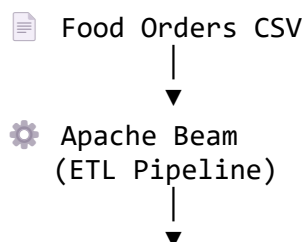
A food delivery company receives hundreds of customer orders every day from multiple restaurants and payment channels.

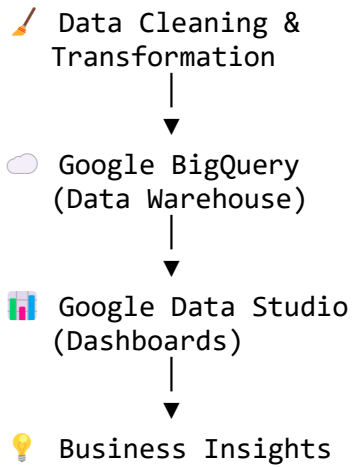
Management wanted answers to several business questions:

- How many orders are successfully delivered?
- What payment methods are customers using?
- Which restaurants generate the most customer engagement?
- What are customers saying about the service?
- What factors are driving poor customer ratings?
- How can customer satisfaction be improved?

The challenge was that the data existed as a raw CSV file and required cleaning, transformation, cloud storage, and reporting before any meaningful insights could be extracted.

Solution Architecture





Technology Stack

- Python
 - Apache Beam
 - Google Cloud Platform (GCP)
 - Google BigQuery
 - Google Data Studio
 - SQL
 - Cloud Storage
-

Dataset

The dataset contains food delivery transactions with:

- Customer ID
- Order Date
- Timestamp
- Order ID
- Ordered Item
- Amount
- Payment Mode
- Restaurant
- Delivery Status
- Customer Rating
- Customer Feedback

My Approach

Rather than analysing the CSV file directly, I designed a cloud analytics workflow similar to what many organisations use in production environments.

The objective was to:

1. Ingest raw transactional data.
2. Clean and standardise records.
3. Apply business logic and classification.
4. Store processed data in a cloud data warehouse.
5. Create dashboards for operational and customer experience reporting.
6. Generate actionable business insights.

This project allowed me to combine Data Engineering, Cloud Analytics, and Business Intelligence into a single solution.

ETL Pipeline Development

Step 1: Data Ingestion

The Apache Beam pipeline reads the raw CSV file containing food order transactions.

Input file:

`food_daily_10_2020.csv`

Step 2: Data Cleaning

Several data quality issues were identified in the source data.

Remove trailing punctuation

Example:

Chow?Mein:

became:

Chow Mein

Remove unwanted special characters

Removed:

?
%
&

Standardise text

Converted records to lowercase to improve consistency and prevent duplicate category values.

Example:

Cash

became:

cash

Add processing flag

An additional processing indicator column was added for downstream tracking.

Step 3: Data Transformation

The pipeline separated records into two business categories.

Delivered Orders

`status = delivered`

Other Status Orders

`status != delivered`

This allowed delivery performance to be analysed independently from unsuccessful orders.

Step 4: Data Validation

During execution the pipeline validated record counts.

Total Orders

891

Delivered Orders

869

Other Status Orders

22

This ensured all records were processed successfully.

Cloud Data Warehouse

The transformed data was loaded into Google BigQuery.

Dataset:

`dataset_food_orders_latest`

Tables:

`delivered_orders`

`other_status_orders`

Reporting View:

`daily_food_orders`

This architecture separates processing from reporting and follows modern cloud analytics practices.

Dashboard Development

Two dashboards were developed using Google Data Studio.

Dashboard 1: Daily Operations Dashboard

KPIs

Total Orders

869

Average Customer Rating

2.9 / 5

Delivery Success Rate

97.5%

Visualisations

Payment Mode Distribution

Analysed customer payment behaviour across:

- Wallet
- Card
- Online
- Cash

Customer Ratings by Restaurant

Compared customer satisfaction across restaurants.

Daily Dashboard Insights

Wallet is the most preferred payment mode

Customers demonstrated a strong preference for digital payment methods.

Strong delivery performance

The delivery operation achieved a 97.5% success rate, indicating efficient fulfilment processes.

Customer satisfaction opportunity

Despite strong delivery performance, average customer ratings remained at 2.9 / 5, suggesting customer experience improvements are required.

Highest customer engagement

Delhi Pot and Others generated the highest volume of customer ratings and engagement.

Dashboard 2: Monthly Customer Experience Dashboard

The monthly dashboard focused on understanding customer sentiment and service quality.

Visualisations

Monthly Rating Distribution

Analysed overall customer rating behaviour.

Customer Feedback Analysis

Identified recurring customer complaints and service issues.

Monthly Dashboard Insights

Most common rating

2-Star ratings were the most common customer response.

This indicates that customer satisfaction remains a business challenge.

Top Customer Complaint

Stale Food

This emerged as the most frequently reported issue.

Additional Customer Concerns

- High Price
 - Difficult Ordering Process
 - Late Delivery
 - Coupon Issues
-

Business Recommendations

Based on the analytics findings, the company should focus on:

Food Quality Improvements

Reduce stale food complaints through stronger quality control and operational monitoring.

Pricing Strategy Review

Investigate pricing concerns and improve perceived value.

Ordering Experience Optimisation

Simplify ordering workflows and improve customer usability.

Delivery Process Improvements

Reduce delivery delays and improve communication with customers.

Challenges Encountered & Problem Solving

One of the most valuable aspects of this project was solving multiple real-world cloud and pipeline implementation issues.

Challenge 1: BigQuery Location Mismatch

Problem

Dataset creation failed due to an unsupported location configuration.

Solution

Configured the dataset region correctly using:

```
dataset.location = "australia-southeast1"
```

This aligned BigQuery resources with the project region.

Challenge 2: Project ID Configuration

Problem

The pipeline attempted to connect to an incorrect Google Cloud project.

Solution

Validated and configured the correct project settings using Google Cloud Shell.

Challenge 3: Python 2 to Python 3 Migration

Problem

The original script was written in legacy Python 2 syntax.

Example:

```
print row
```

Solution

Migrated all code to Python 3 syntax.

Example:

```
print(row)
```

This ensured compatibility with modern environments.

Challenge 4: Apache Beam Compatibility Issues

Problem

Older Beam examples generated compatibility and execution issues.

Solution

Updated execution logic and modified pipeline configuration to work with current Apache Beam versions.

Challenge 5: Cloud Storage Bucket Setup

Problem

Pipeline execution required temporary cloud storage for processing.

Solution

Created and configured a dedicated Cloud Storage bucket and supplied a valid temporary location during execution.

Challenge 6: Pipeline Execution Errors

Problem

Several runtime errors occurred during development.

Solution

Debugged:

- Pipeline options
- Temporary storage locations
- BigQuery table references
- Data schemas

until successful execution was achieved.

Challenge 7: BigQuery Connectivity Issues

Problem

BigQuery write operations initially failed.

Solution

Validated:

- APIs
- Permissions
- Dataset creation
- Table configuration

before successfully loading transformed data.

Challenge 8: Google Data Studio Setup

Problem

Connecting cloud data to reporting dashboards required additional configuration.

Solution

Configured:

- BigQuery connectors

- Calculated fields
- KPI cards
- Dashboard layouts
- Visualisations

to build reporting-ready dashboards.

Key Results

Metric	Result
Total Orders	891
Delivered Orders	869
Other Orders	22
Delivery Success Rate	97.5%
Average Customer Rating	2.9 / 5
Preferred Payment Method	Wallet
Most Common Rating	2 Stars
Top Customer Complaint	Stale Food

Skills Demonstrated

Data Engineering

- Apache Beam
- ETL Pipeline Development
- Data Cleaning
- Data Transformation
- Data Validation

Cloud Analytics

- Google Cloud Platform
- BigQuery
- Cloud Storage

Business Intelligence

- Google Data Studio
- Dashboard Development

- KPI Design
- Data Storytelling

Analytics

- Customer Satisfaction Analysis
 - Delivery Performance Analysis
 - Feedback Analysis
 - Business Insight Generation
-

Project Outcome

This project successfully transformed raw food delivery transaction data into a cloud-based analytics solution capable of generating actionable business insights.

The solution demonstrates a complete analytics workflow from data ingestion through reporting and highlights practical experience in:

- Data Engineering
- Cloud Analytics
- BigQuery
- Business Intelligence
- Problem Solving
- Data Storytelling

while delivering measurable insights that can support operational and customer experience improvements.

This project strengthened my understanding of cloud analytics architectures and provided hands-on experience with Apache Beam, BigQuery, and Google Data Studio in a real-world business scenario.